FX SCALPING BUILD PLAN — integrations, training data, profitability

Session 25 · 2026-10-27 · status: owner-requested deep dive (supersedes the wide-range 30–150 pip plan for FX) Scope: the 7 FX pairs on the IC Markets trunk (EUR/USD, GBP/USD, USD/JPY, AUD/USD, USD/CAD, USD/CHF, USD/SGD) traded as **non-LLM scalping cells**. LLMs never in the hot path (R2/R3). Every number below is either *measured*, *web-verified with source*, or marked *unverified*.

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§1 · The cost math — what a scalper actually pays

Measured on the IC Markets MT Raw account (26-Oct): **spread 0.32 pip avg + commission $7/lot round-turn = 0.70 pip**.

Path	Cost per round trip	Formula
Taker (market order both ways)	**1.02 pips**	0.32 spread + 0.70 commission
Maker (limit both ways)	**0.70 pips**	commission only — you *earn* the spread if both fill
Alt: cTrader $3/$100k/side	0.92 pips taker	0.32 + 0.60

Breakeven win rate = `50% + friction ÷ (2 × target)`:

Target	Taker (1.02)	Maker (0.70)	Verdict
1 pip	**101%**	85%	❌ mathematically dead on taker
2 pips	75.5%	67.5%	❌
3 pips	**67.0%**	61.7%	❌ brutal
5 pips	**60.2%**	57.0%	⚠️ hard
10 pips	55.1%	53.5%	⚠️ reachable
20 pips	52.6%	51.8%	✅

Two conclusions from the table alone:

1. **Maker on entry is mandatory.** Taker scalping under 5 pips is a losing game before skill enters. Standing rule for E6: *enter on limit orders (earn the spread); market orders are for exits only.*
2. **Real scalping band = 3–10 pips**, requiring **53–67% win rate**. Not 1–2 pips.

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§2 · Latency budget — measured tiers

Tier	Latency	Slippage cost	Verdict
Ideal	**< 5 ms**	~0.01 pip	NY4 VPS → IC = **< 1 ms** ✅ achievable
Excellent	< 30 ms	0.01–0.1 pip	✅
Hard ceiling	**< 100 ms**	grows fast	⚠️ at 100 ms a 5-pip scalp loses ~0.2 pip/trade ≈ 4% of profit, compounding over hundreds of trades/month
Home MT4	500–700 ms	—	❌ enters *after* the move
News (NFP/CPI/FOMC)	**sub-10 ms required**	pips/second	⚠️ only after NY4 is proven

Source: vpsforextrader.com "Scalping Execution Latency 2026: Every Component Measured" (2026-04-05).

Implication: the **NY4 VPS is not optional for scalping — it is build step 1.** Without it, do not build this plan.

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§3 · Feature set — what the scalpers train on

Group D+ · microstructure (the core, from the cTrader feed)

Feature	Source	Why
Microprice (size-weighted bid/ask mid)	Level II	leads the mid — best short-horizon signal class
Order flow imbalance (OFI)	Level II	Cont-Kukanov-Stoikov (2014): explains a large share of next-tick price change
Book imbalance	Level II	EFMA 2025: predictive up to **30 s lag**; midpoint at 100 ms / 1 s / 2 s horizons
Spread regime state ⭐	ticks	#1 cost feature — spread widens at news and session opens *exactly when scalpers trade*
Tick count (hourly) = volume proxy	ticks	~90% corr with CME futures volume at hourly aggregation — **labelled proxy, never "real volume"** (R8: FX has no live volume tape)
Session clock (Sydney→Tokyo→London→NY)	computed	liquidity by hour
Realized vol / ATR regime	computed	skip when friction > edge
IC liquidity-pool depth	Level II	⚠️ verify volume-bearing vs synthetic at onboarding — label "broker book"

Group A/B/C · macro filters (slower clock, from free PIT-safe sources)

Feature	Source	Use in scalping
Calendar + surprise index	FXStreet API / FRED	**trade filter only** — no entries in the ±5 min window (spread blows out)
Rate differentials, real yields	FRED (**vintages** = lookahead killer)	session bias direction
COT net positioning	CFTC API (works without token, free)	daily regime — which side not to fade

⚠️ Honest academic caveats

Evidence	What it says
Kolm 2023, *Mathematical Finance*	models trained on **order flow** significantly outperform models trained directly on order books
NSE OIB study (ScienceDirect 2021)	order-imbalance predictability strong in the **first 5 minutes, dead by 30 minutes**
Imbalance effect is **stronger for larger ticks**	**FX majors have the smallest ticks of any asset class → the weakest raw imbalance signal**

Translation: the edge on FX majors will be *smaller* than the same strategy on NSE stocks or futures. Expectation stays modest — **spread-regime + session clock + maker spread capture carry more weight than raw book imbalance.**

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§4 · Data integration — cTrader Open API (open decision O2 = RESOLVED here)

Need	cTrader message	Notes
Live bid/ask ticks	`ProtoOASubscribeSpotsReq` → `ProtoOASpotEvent`	price = raw ÷ 100000, round to `digits`
Level II depth	`ProtoOASubscribeDepthQuotesReq` → `ProtoOADeathEvent` / `ProtoOADepthEvent`	fires on every book change — OFI/microprice source
Historical ticks	`ProtoOAGetTickDataReq` (BID/ASK type)	backtest fuel
Symbol master	`digits, pipPosition, minVolume, maxVolume, stepVolume, lotSize, swapLong, swapShort`	swap pulled from API, never hardcoded
Connection	TCP **or** WebSocket, Protobuf/JSON, `live.ctraderapi.com:5035`, AWS Global Accelerator	Linux-native ✅
SDK	`pip install ctrader-open-api` (Python/Twisted), Go, TypeScript	
Rate limits	**5 req/s historical · 50 req/s live**	tick-history pull = days, not hours
FIX 4.4	no minimum deposit or volume	

Why cTrader and not MT5: IC Markets — "We do not offer a FIX API connection into MetaTrader 4 or 5. You can, however, connect to our cTrader platform via FIX API." MT5 automation is Windows-only → risk R1 avoided.

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§5 · Backtest engine — tick-level or it's worthless

Candle backtests **cannot** test scalping. Required additions to E4:

Requirement	Why
Tick-by-tick replay	intrabar sequencing decides 3-pip wins
Queue position modeling	maker fills: you join the back of the queue — fill probability ≠ 1
Partial fills	your size vs displayed depth
Spread as a time series	0.32 avg hides 2–5 pip news spikes
Latency injection	run the sim at 1 / 10 / 100 ms — watch the strategy die at 100
Slippage from **own E7 fills**	live-measured, not assumed
CPCV + DSR gates	existing E4 standard — no random splits on autocorrelated ticks

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§6 · Training pipeline

Step	What	Gate
1	Pull tick + Level II history (5 req/s — **start first**, it is the long pole)	coverage log per pair
2	Assemble PIT panel: microstructure + spread regime + session clock + macro filters	0 lookahead rows
3	**Label** = forward 3–10 pip move, **net of 1.02 taker / 0.70 maker + measured slippage**	labels net, never gross
4	Train E3 (GBM / small NN → ONNX; SB3 baseline)	seed-invariance: same seed ⇒ same weight hash, twice
5	CPCV + DSR + crisis replays (news-spike kits)	DSR ≥ 1

Step	What	Gate
6	**Spread-regime audit** — performance *by spread bucket*	positive in tight bucket only; wide bucket flat = not traded
7	Latency sweep: same model at 1 / 10 / 100 ms	edge survives at real venue latency
8	Shadow 3–6 months, zero orders	**net expectancy > 0 after real friction**
9	Live 1 pair (EUR/USD), maker-entry rule enforced in E6	5 sessions ledger-matched
10	Expand → 7 pairs	governor caps hold

Profitability bar (honest, stated once):

> **Win rate > 55% (target 60%) at 3–10 pip targets · profit factor > 1.2 (target 1.5) · max drawdown < 20% (target 10%) · DSR ≥ 1 · order latency < 100 ms (target < 50 ms) · net expectancy > 0 after 1.02-pip friction + measured slippage — over ≥ 3 months out-of-sample shadow.** Anything short of that does not ship. Profit is never promised; only this bar is measurable.

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§7 · Build order

#	Step	Owner input	Gate
1	**NY4 VPS** for the FX hub	buy (~$100–300/mo)	ping to IC < 1 ms
2	**cTrader account + API keys** (O2 resolved)	your login	read-only quote stream works
3	Start **historical tick + depth pull** immediately	none	5 req/s rate limit = days of pull
4	Symbol master: resolve Brent symbol (`XBRUSD` / `XBRUSH` / `BRENT`), **confirm DOM volume-bearing vs synthetic**	none	exactly 1 symbol each; DOM labelled
5	Feature engine E1: microstructure group wired	none	offline/online bit-exact in CI
6	Tick backtest engine in E4 (queue / partial fills / spread series / latency injection)	none	matches cTrader fills on a sample window
7	Train first cell — EUR/USD scalp	none	CPCV + DSR + spread-bucket audit pass
8	Shadow ≥ 3 months	none	**net expectancy > 0 after friction**
9	Live EUR/USD only, maker-entry rule	sign-off	5 sessions ledger-matched
10	Expand → 7 FX pairs	sign-off per tranche	E5 caps hold

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§8 · Risk register

ID	Risk	Sev	Mitigation
S1	1.02-pip taker friction is dead under 5-pip targets	**High**	**maker-entry rule enforced in E6** (0.70 path)
S2	FX majors = smallest ticks → weakest imbalance signal	**High**	weight spread-regime + session clock; modest expectations
S3	Spread widens at news/session opens exactly when scalping	**High**	spread-regime = hard gate, no entry in widen state
S4	Latency > 100 ms kills the edge	**High**	NY4 mandatory (step 1); latency sweep in training

ID	Risk	Sev	Mitigation
S5	Queue position — maker orders may not fill	**Med**	fill-probability model in backtest; capped chase rules
S6	Adverse selection — IC is the CFD issuer, sees your flow	**Med**	no one-way quoting, modest size, E7 slippage feedback loop
S7	Tick-history pull slow (5 req/s)	**Med**	start at build step 3 — long pole of the plan
S8	Lookahead from revised news prints	**Med**	FRED vintages + as-of stamps enforced in CI
S9	No real FX volume tape (R8)	**Med**	hourly tick-count proxy + optional CME cross-check, labelled proxy

§9 · Could not verify (needs the live account)

Item	Status
IC DOM carries real volumes or is synthetic	onboarding step 4
Brent symbol — `XBRUSD` / `XBRUSH` / `BRENT`	same
Full swap schedule per pair (XAUUSD long −$53.31/lot/night measured; EUR/USD unmeasured)	pull from cTrader symbol master
Historical depth availability via API (docs only show *live* depth subscribe)	test at step 3 — if no history, record live going forward
CME FX futures live volume	`cme.com` HTTP 403 IP-blocked → paid feed; optional cross-check only

§10 · Provenance — measured vs verified vs unverified

Claim	Class
Spread 0.32 pip (EUR/USD, 26-Oct), commission $7/lot RT = 0.70 pip, round trip 1.02 pips	**Measured**
Breakeven win-rate table (§1)	**Derived** from measured friction
Latency tiers and slippage ranges (§2)	**Web-verified** — vpsforextrader.com 2026-04-05
OFI / book-imbalance predictive evidence, Kolm 2023, tick-size caveat (§3)	**Web-verified** — Cont-Kukanov-Stoikov 2014, EFMA 2025, NSE OIB 2021, Kolm *Mathematical Finance* 2023
cTrader messages, rate limits, endpoint, SDK (§4)	**Web-verified** — help.ctrader.com Open API docs + IC Markets support statement
Tick-count ≈ 90% corr with CME at hourly	**Web-verified** (Marney/Dxa) — correlation must be **re-measured on our 7 pairs**
Win-rate / PF / DD / latency targets (§6)	**Targets**, not results
Actual profitability	**Unverified — cannot be promised.** Only the measurable bar ships

Artifact of Session 25. O2 (connectivity) resolved here = cTrader Open API. Supersedes the wide-range 30–150 pip FX plan for the FX trunk; the wide-range styles (session/carry/news-fade) remain valid as future second styles per O5.